

HumanEval typicality metrics — qwen3.5-9b & gemma-4-31b-it (2026-06-16)

Settings: s2 = RankAlign; s4 = New+fsx+self-tc (FLORA-PMI); base = untrained qwen. Eval: basetyp = self-typ + base-typ; self = self-typ no-base; basetypneg = neg-typ + base-typ; neg = neg-typ no-base. Gen variant = tc (typicality-corrected). Metrics averaged over the 82 humaneval test tasks (gemma multi-s4-basetyp: 70, see /datastor1-full note).

Table 1: HumanEval typicality metrics (gen variant = tc), x100: gen ROC, val ROC, val acc, Pearson(gen,val).

Model	Data	Setting	Eval	gen ROC	val ROC	val acc	Pearson
gemma-4-31b-it (trained)	multi	s2	basetyp	90.7	90.0	76.5	64.9
gemma-4-31b-it (trained)	multi	s2	basetypneg	94.2	90.0	76.5	70.7
gemma-4-31b-it (trained)	multi	s2	neg	82.4	90.0	76.5	53.2
gemma-4-31b-it (trained)	multi	s2	self	85.6	90.0	76.5	58.3
gemma-4-31b-it (trained)	multi	s4	basetyp	90.1	94.4	82.6	69.0
gemma-4-31b-it (trained)	multi	s4	self	84.0	94.4	82.6	57.6
gemma-4-31b-it (trained)	upper	s2	basetyp	86.9	91.4	76.8	41.0
gemma-4-31b-it (trained)	upper	s2	basetypneg	88.2	91.4	76.8	55.5
gemma-4-31b-it (trained)	upper	s2	neg	86.3	91.4	76.8	59.9
gemma-4-31b-it (trained)	upper	s2	self	85.2	91.4	76.8	41.2
gemma-4-31b-it (trained)	upper	s4	basetyp	90.7	93.1	84.2	63.3
gemma-4-31b-it (trained)	upper	s4	self	84.3	93.1	84.2	52.5
qwen3.5-9b (trained)	multi	s2	basetyp	93.1	84.8	77.9	64.7
qwen3.5-9b (trained)	multi	s2	basetypneg	96.5	84.8	77.9	70.1
qwen3.5-9b (trained)	multi	s2	neg	80.3	84.8	77.9	75.2
qwen3.5-9b (trained)	multi	s2	self	90.2	84.8	77.9	61.3
qwen3.5-9b (trained)	multi	s4	basetyp	89.8	89.1	80.5	66.0
qwen3.5-9b (trained)	multi	s4	self	82.9	89.1	80.5	56.5
qwen3.5-9b (trained)	upper	s2	basetyp	87.2	83.3	74.3	55.3
qwen3.5-9b (trained)	upper	s2	basetypneg	90.2	83.2	74.3	69.5
qwen3.5-9b (trained)	upper	s2	neg	79.3	83.3	74.3	68.9
qwen3.5-9b (trained)	upper	s2	self	82.2	83.3	74.3	46.0
qwen3.5-9b (trained)	upper	s4	basetyp	83.2	86.3	76.4	58.9
qwen3.5-9b (trained)	upper	s4	self	78.6	86.3	76.5	51.5
qwen3.5-9b (BASE)	multi	base	neg	59.9	83.5	67.0	29.3
qwen3.5-9b (BASE)	multi	base	self	69.0	83.4	66.8	27.2
qwen3.5-9b (BASE)	upper	base	neg	65.7	81.7	65.3	30.4
qwen3.5-9b (BASE)	upper	base	self	69.1	81.7	65.3	26.5